

Brief introduction to conformal prediction, with a discussion for computer experiments

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Outline

Context, introduction to conformal prediction

Resampling and training conditional coverage

Distribution shift

Adaptivity and test conditional coverage

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Context, introduction to conformal prediction

Resampling and training conditional coverage

Distribution shift

Adaptivity and test conditional coverage

Why prediction intervals?

- ▶ **Just providing model predictions is no longer sufficient** in industrial and high-stake fields
- ▶ Need for **prediction intervals** around the predictions, which should contain the true but unobserved value of the target with a certain level of confidence
- ▶ Expected benefits
 - ▶ *Trust* with end-users of the ML model, since they will be able to see how confident it is when making a prediction
 - ▶ Opening the path for *sequential or adaptive design of experiments*, by requesting new labeled data in the feature regions where the model is not confident (a.k.a. *active learning*)
 - ▶ Detecting potential out-of-distribution data, if the intervals are unexpectedly wide

This is not a new problem

It is quite easy to generate some kind of intervals for some models, with different underlying ideas:

- ▶ Explicit central limit theorems that are available for some models (polynomial regression, local-averaging methods, ...)
- ▶ Quantile regression, where the model is trained to specifically learn quantiles of the target conditional distribution instead of the mean
- ▶ Bayesian paradigm (Gaussian processes or Bayesian neural networks more recently, ...)
- ▶ Resampling methods (bootstrap, cross-validation, leave-one-out, jackknife, ...)
- ▶ Heuristic approaches, lot of popularity in the neural network community (deep ensembles, drop-out, ...)

Notations & setup

- ▶ **Supervised learning setting:** dataset $\mathcal{D}_n = (\mathbf{X}_i, Y_i)_{i=1, \dots, n}$ from $P_{\mathbf{X}Y}$ defined on product space $\mathcal{X} \times \mathcal{Y}$
- ▶ **Focus here on regression:**
 - ▶ $\mathbf{X} = (X^{(1)}, \dots, X^{(d)}) \in \mathcal{X} \subset \mathbb{R}^d$ explanatory variables (*features*)
 - ▶ $Y \in \mathcal{Y} \subset \mathbb{R}$ *target* variable
- ▶ **Predictor:** $\hat{\mu}_{\mathcal{D}_n} : \mathbb{R}^d \rightarrow \mathbb{R}$ is the prediction function.

Our objective: formal coverage guarantees

We aim to build a distribution-free prediction interval defined as follows:

Definition (Prediction interval/band)

A **prediction interval** $\hat{\mathcal{C}}_{\mathcal{D}_n}$ with error level $\alpha \in (0, 1)$ is a function

$$\hat{\mathcal{C}}_{\mathcal{D}_n} : \mathcal{X} \rightarrow \{\text{subsets of } \mathcal{Y}\}$$

built from an i.i.d. sample $\mathcal{D}_n = (\mathbf{X}_i, Y_i)_{i=1, \dots, n}$ from $P_{\mathbf{X}Y}$ such that, for a new i.i.d. pair $(\mathbf{X}_{n+1}, Y_{n+1}) \sim P_{\mathbf{X}Y}$, we have

$$\mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}(\mathbf{X}_{n+1}) \right) \geq 1 - \alpha, \quad (1)$$

where **the probability is over all data** $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$.

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where **the probability is over all data** $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$.

- ▶ **Marginal coverage:** The probability is marginalized over all data randomly drawn for training and testing
- ▶ **Training conditional coverage:** $\mathbb{P} (Y_{n+1} \in \hat{C}_{\mathcal{D}_n}(\mathbf{X}_{n+1}) | \mathcal{D}_n) \geq 1 - \alpha$.

In practice, training conditional coverage is a must-have: we want guarantees for the specific training sample we have at hand.

Our objective: formal coverage guarantees

From this definition, notice that we target two specific properties for a prediction interval:

1. It must be **distribution-free**, i.e. the coverage guarantee must hold without assumptions on the data generating process $P_{\mathcal{X}\mathcal{Y}}$
2. It must be valid in a **non-asymptotic** framework, i.e. for any number n of samples used to build it

This is exactly what conformal prediction does, as we will see in what follows.

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Remark

Observe that in the definition above:

- ▶ Probability is over all data, this means that coverage is guaranteed in average over all random draws of training data (used to build the prediction interval) and random draws of testing data (where we predict)
- ▶ Testing data is supposed here to follow the same distribution as the training data, meaning that we will have to make adjustments to handle important practical situations (timeseries, active learning)

A central ingredient for conformal prediction is the concept of exchangeable random variables.

Definition (Exchangeability)

Random variables $Z_1, \dots, Z_n$ are **exchangeable** if

$$(Z_1, \dots, Z_n) \stackrel{d}{=} (Z_{\sigma(1)}, \dots, Z_{\sigma(n)})$$

for every finite permutation σ of the indices $1, \dots, n$.

Preliminaries on exchangeability

Four important practical cases for us lead to exchangeable sequences:

Proposition (Particular cases of exchangeable sequence)

The random variables $Z_1, \dots, Z_n$ are exchangeable in the following cases:

- (i) $Z_1, \dots, Z_n$ are i.i.d. from some distribution P
- (ii) $Z_1, \dots, Z_n | \theta$ are i.i.d. from some distribution $P(\cdot | \theta)$ indexed by some random vector $\theta \sim P_\theta$ (*conditional i.i.d*)
- (iii) $Z_1, \dots, Z_n$ are sampled uniformly without replacement from a finite set
- (iv) $Z_1 = f(W_1), \dots, Z_n = f(W_n)$ with $W_1, \dots, W_n$ exchangeable and f any function

Proposition (A simple but useful result on quantiles)

For any $z_1, \dots, z_n, t \in \mathbb{R}$ and quantile level $\tau \in [0, 1]$, we have

$$t \leq \text{Quantile}_\tau(z_1, \dots, z_n, t) \Leftrightarrow t \leq \text{Quantile}_{\tau \frac{n+1}{n}}(z_1, \dots, z_n)$$

Proposition (A central result for exchangeable sequences)

If $Z_1, \dots, Z_n$ are exchangeable, then $\forall i = 1, \dots, n$

$$\mathbb{P}(Z_i \leq \text{Quantile}_\tau(Z_1, \dots, Z_n)) \geq \tau$$

for any quantile level $\tau \in [0, 1]$.

The simplest CP method: split conformal prediction

We typically want a prediction interval centered around a point prediction given by $\hat{\mu}_{\mathcal{D}_n}$, which means that we seek a prediction interval of the form

$$\hat{\mu}_{\mathcal{D}_n}(\mathbf{X}_{n+1}) \pm \dots$$

that is for example

$$\hat{C}_{\mathcal{D}_n}(\mathbf{x}) = [\hat{\mu}_{\mathcal{D}_n}(\mathbf{x}) - \hat{q}_n, \hat{\mu}_{\mathcal{D}_n}(\mathbf{x}) + \hat{q}_n]$$

or

$$\hat{C}_{\mathcal{D}_n}(\mathbf{x}) = \{y \in \mathbb{R} : |y - \hat{\mu}_{\mathcal{D}_n}(\mathbf{x})| \leq \hat{q}_n\}$$

for some $\hat{q}_n$ to be determined.

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Using the last equality, the targeted coverage is thus

$$\mathbb{P}(Y_{n+1} \in \hat{C}_{\mathcal{D}_n}(\mathbf{X}_{n+1})) = \mathbb{P}(|Y_{n+1} - \hat{\mu}_{\mathcal{D}_n}(\mathbf{X}_{n+1})| \leq \hat{q}_n) = \mathbb{P}(R_{n+1} \leq \hat{q}_n) \geq 1 - \alpha$$

where we denote $R_i = |Y_i - \hat{\mu}_{\mathcal{D}_n}(\mathbf{X}_i)|$ the absolute residuals for $i = 1, \dots, n+1$.

The simplest CP method: split conformal prediction

As is done in supervised learning, we can split the training set $\mathcal{D}_n = \mathcal{D}_n^{\text{pttrain}} \cup \mathcal{D}_n^{\text{cal}}$ in two disjoint sets:

- ▶ $\mathcal{D}_n^{\text{pttrain}}$ is the *proper training* set used to build the predictor $\hat{\mu}_{\mathcal{D}_n^{\text{pttrain}}}$ only (in CP it may sometimes be referred to as the *pretraining* set)
- ▶ $\mathcal{D}_n^{\text{cal}}$ is the hold-out *calibration set* on which the residuals are computed, which has the same role as the validation set in supervised learning

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By computing the quantile of the residuals on $\mathcal{D}_n^{\text{cal}}$, we can achieve the required coverage.

To see this, let us consider the absolute residuals $R_i = |Y_i - \hat{\mu}_{\mathcal{D}_n^{\text{pttrain}}}(\mathbf{X}_i)|$ for $i \in \mathcal{D}_n^{\text{cal}}$ on the calibration set and R_{n+1} the test residual.

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Of course these residuals are not independent since they all depend on $\hat{\mu}_{\mathcal{D}_n^{\text{pttrain}}}$ and thus on $\mathcal{D}_n^{\text{pttrain}}$. However, when we condition on $\mathcal{D}_n^{\text{pttrain}}$, they obviously become i.i.d., meaning that **they are exchangeable** by Proposition 1 (ii).

Split conformal prediction

This leads directly to the following result:

$$\mathbb{P} \left(R_{n+1} \leq \text{Quantile}_{(1-\alpha) \frac{n_{\text{cal}}+1}{n_{\text{cal}}}} (R_i, i \in \mathcal{D}_n^{\text{cal}}) \right) \geq 1 - \alpha \quad (2)$$

or

$$\mathbb{P} (Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{split}}(\mathbf{X}_{n+1})) \geq 1 - \alpha$$

for a prediction interval of the form

$$\hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{split}}(\mathbf{x}) = \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}(\mathbf{x}) \pm \hat{q}_{\mathcal{D}_n^{\text{cal}}}$$

where

$$\hat{q}_{\mathcal{D}_n^{\text{cal}}} = \text{Quantile}_{(1-\alpha) \frac{n_{\text{cal}}+1}{n_{\text{cal}}}} (R_i, i \in \mathcal{D}_n^{\text{cal}}) = \hat{q}_{n_{\text{cal}}, \alpha}^+ \{R_i, i \in \mathcal{D}_n^{\text{cal}}\}$$

and n_{cal} is the size of the calibration set.

This is called **split conformal prediction**, *split* obviously referring to the fact that we separated the training set in a proper training set and a calibration set.

Theorem (Coverage for split conformal - Vovk et al. (2005))

If $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$ are exchangeable, the split conformal interval satisfies

$$\mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{split}}(\mathbf{X}_{n+1}) \right) \geq 1 - \alpha.$$

Marginal vs training conditional coverage

The previous coverage guarantee is **marginal**.

On the other hand, we may look at the coverage where we only marginalize over the test set, i.e.

$$\mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{split}}(\mathbf{X}_{n+1}) | \mathcal{D}_n \right),$$

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Indeed, we can show that for split conformal

$$\mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{split}}(\mathbf{X}_{n+1}) | \mathcal{D}_n \right) \sim \text{Beta}(l, n_{\text{cal}} + 1 - l)$$

where $l = \lceil (1 + n_{\text{cal}})(1 - \alpha) \rceil$ by classical results on the distribution of order statistics (under the assumption that there are no ties in the residuals), see Angelopoulos & Bates (2021) or Tibshirani (2024) for illustrations.

Marginal vs training conditional coverage

We thus deduce that the moments of the training conditional coverage satisfy

$$\begin{aligned}\mathbb{E} \left\{ \mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{split}}(\mathbf{X}_{n+1}) | \mathcal{D}_n \right) \right\} &= \frac{[(1+n_{\text{cal}})(1-\alpha)]}{1+n_{\text{cal}}} \in \left[1-\alpha, 1-\alpha + \frac{1}{n_{\text{cal}}+1} \right] \\ \text{Var} \left\{ \mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{split}}(\mathbf{X}_{n+1}) | \mathcal{D}_n \right) \right\} &= \frac{l(n_{\text{cal}}+1-l)}{(n_{\text{cal}}+1)^2(n_{\text{cal}}+2)} \approx \frac{\alpha(1-\alpha)}{n_{\text{cal}}+2}\end{aligned}$$

where the last approximation is valid when n_{cal} is sufficiently large.

This means that n_{cal} , the number of calibration samples, plays a major role on how close the training conditional coverage is to its mean, which itself is close to $1-\alpha$.

Marginal vs training conditional coverage

More precisely, a more quantitative result states that (Vovk 2012)

$$\mathbb{P} \left\{ \mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{split}}(\mathbf{x}_{n+1}) | \mathcal{D}_n \right) \geq 1 - \alpha - \sqrt{\frac{\log(1/\delta)}{2n_{\text{cal}}}} \right\} \geq 1 - \delta.$$

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- ▶ All in one, n_{cal} should be chosen as large as possible in order to have a training conditional coverage close to $1 - \alpha$ (but remember that marginal coverage will hold for any value of n_{cal})
- ▶ However, this means that n_{ptrain} the size of the proper training set will then be smaller, thus leading to a degraded prediction function $\hat{\mu}_{\mathcal{D}_n^{\text{pttrain}}}$, and ultimately to larger and thus less informative prediction intervals in general

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Split conformal prediction fundamentally comes with such a trade-off, and we will see later other variants which overcomes this limitation (at the cost of additional assumptions, or other types of limitations).

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Beyond split CP: jackknife+ and CV+

Split CP sacrifices training data: resampling methods retain data but modify the interval construction using leave-one-out (LOO, i.e. jackknife) or cross-validation (CV) predictors.

Jackknife+ interval: uses LOO predictions $\hat{\mu}_{-i}$ and residuals

$$R_i^{\text{LOO}} = |Y_i - \hat{\mu}_{-i}(\mathbf{X}_i)|:$$

$$\hat{C}_{\mathcal{D}_n}^{\text{jack}+}(\mathbf{x}) = [\hat{q}_{n,\alpha}^{-}\{\hat{\mu}_{-i}(\mathbf{x}) - R_i^{\text{LOO}}\}, \hat{q}_{n,\alpha}^{+}\{\hat{\mu}_{-i}(\mathbf{x}) + R_i^{\text{LOO}}\}]$$

CV+ Interval: uses K -fold CV predictions $\hat{\mu}_{-\mathcal{D}_k}$ and residuals R_i^{CV} :

$$\hat{C}_{\mathcal{D}_n}^{\text{CV}+}(\mathbf{x}) = [\hat{q}_{n,\alpha}^{-}\{\hat{\mu}_i^{\text{CV}}(\mathbf{x}) - R_i^{\text{CV}}\}, \hat{q}_{n,\alpha}^{+}\{\hat{\mu}_i^{\text{CV}}(\mathbf{x}) + R_i^{\text{CV}}\}]$$

Definition (Shortcut for quantiles (Barber et al. (2021b)))

- ▶ $\hat{q}_{n,\alpha}^{+}\{z_i\} = z_{\lceil(1-\alpha)(n+1)\rceil}$ the sample $(1 - \alpha)\frac{n+1}{n}$ -quantile of $z_1, \dots, z_n$
- ▶ $\hat{q}_{n,\alpha}^{-}\{z_i\} = z_{\lfloor\alpha(n+1)\rfloor}$, the sample $\alpha\frac{n+1}{n}$ -quantile of $z_1, \dots, z_n$ up to a different rounding

Beyond split CP: jackknife+ and CV+

Theorem (Coverage for jackknife+ - Barber et al. (2021b))

If $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$ are exchangeable and the prediction algorithm is symmetric, the jackknife+ interval satisfies

$$\mathbb{P}(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{jack}+}(\mathbf{X}_{n+1})) \geq 1 - 2\alpha.$$

Theorem (Coverage for CV+ - Barber et al. (2021b))

If $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$ are exchangeable and the prediction algorithm is symmetric, the K -fold CV+ interval satisfies

$$\mathbb{P}(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{CV}+}(\mathbf{X}_{n+1})) \geq 1 - 2\alpha - \min \left\{ \frac{2(1 - 1/K)}{n/K + 1}, \frac{1 - K/n}{K + 1} \right\}.$$

When $K = n$ we recover the jackknife+ result, and get a coverage close to $1 - 2\alpha$ when $K \ll n$ is small. For any value of K , we actually have

$$\mathbb{P}(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{CV}+}(\mathbf{X}_{n+1})) \geq 1 - 2\alpha - \sqrt{2/n}.$$

Beyond split CP: jackknife+ and CV+

CV+ intervals also come with training conditional coverage:

Theorem (Training conditional coverage for CV+ - Bian & Barber (2023))

If $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$ are exchangeable, the K -fold CV+ interval with each fold of size m such that $n = Km$ satisfies

$$\mathbb{P} \left\{ \mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{CV}+}(\mathbf{X}_{n+1}) | \mathcal{D}_n \right) \geq 1 - 2\alpha - \sqrt{\frac{2 \log(K/\delta)}{m}} \right\} \geq 1 - \delta.$$

If the number of samples in each fold m is large, training conditional coverage is close to $1 - 2\alpha$. m here actually plays the same role as n_{cal} in the training conditional guarantee of split conformal.

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If the number of samples in each fold m is large, training conditional coverage is close to $1 - 2\alpha$. m here actually plays the same role as n_{cal} in the training conditional guarantee of split conformal.

Finally, observe that jackknife+ corresponds to $m = 1$: without additional assumptions obtaining a training conditional guarantee in this case seems illusory. In fact Bian & Barber (2023) show that it is not possible, both for the jackknife+ and full conformal prediction.

Beyond split CP: jackknife+ and algorithmic stability

To attain **training conditional coverage** for jackknife+, the underlying prediction algorithm must be stable. In essence, removing one training point should not drastically alter out-of-sample predictions.

Definition (Algorithmic stability - Barber et al. (2021b), Amann et al. (2023), Liang & Barber (2023))

A prediction algorithm $\mathcal{A}$ is (ε_n, ν_n) -stable if the predictor it outputs satisfies $\forall i = 1, \dots, n$

$$\mathbb{P} \{ |\hat{\mu}_{\mathcal{D}_n}(\mathbf{X}_{n+1}) - \hat{\mu}_{-i}(\mathbf{X}_{n+1})| \leq \varepsilon_n \} \geq 1 - \nu_n. \quad (SA_1)$$

A prediction algorithm $\mathcal{A}$ is β_n -stable if the predictor it outputs satisfies $\forall i = 1, \dots, n$

$$\mathbb{E} \{ |\hat{\mu}_{\mathcal{D}_n}(\mathbf{X}_{n+1}) - \hat{\mu}_{-i}(\mathbf{X}_{n+1})| \} \leq \beta_n. \quad (SA_2)$$

Remark (Related stability concepts)

Another related but highly different stability criterion in practice is the *in-sample stability*, as opposed to the *out-sample stability* which is another terminology for the one we defined before.

The "in-sample" qualifier refers to the fact that the stability is measured in average for data used to train both predictors, i.e. the in-sample version of $(\mathcal{SA}_1)$ writes

$$\mathbb{P} \{ |\hat{\mu}_{\mathcal{D}_n}(\mathbf{X}_i) - \hat{\mu}_{-i}(\mathbf{X}_i)| \leq \varepsilon_n \} \geq 1 - \nu_n. \quad (\mathcal{SA}_1^{\text{in}})$$

This is a much stronger assumption than $(\mathcal{SA}_1)$. Barber et al. (2021b) give the example of the k -nearest neighbor predictor, which satisfies $(\mathcal{SA}_1)$ with $\varepsilon_n = 0$, $\nu_n = k/n$ but not $(\mathcal{SA}_1^{\text{in}})$.

Beyond split CP: jackknife+ and algorithmic stability

Defining the useful concept of stability is great, but knowing which prediction algorithms satisfy such properties is crucial in practice:

- ▶ Linear regression is $(\mathcal{SA})$ and $(\mathcal{SA}^{\text{in}})$ stable unless $d \approx n$ when it is neither
- ▶ Ridge regression (linear or kernel) is $(\mathcal{SA})$ and $(\mathcal{SA}^{\text{in}})$ stable (Bousquet & Elisseeff 2002), the higher the regularization the more stable
- ▶ Lasso regression is $(\mathcal{SA})$ but not $(\mathcal{SA}^{\text{in}})$ stable (Xu et al. 2011)
- ▶ Bagging of any base predictor is $(\mathcal{SA})$ (Soloff et al. 2024)
- ▶ Random forests are $(\mathcal{SA})$ (Wang et al. 2023)

Beyond split CP: jackknife+ and algorithmic stability

Theorem (Marginal coverage $1 - \alpha$ for jackknife+ - Barber et al. (2021b))

If $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$ are exchangeable, under Assumptions $(\mathcal{AC}^e)$, $(\mathcal{SA}_1)$ and for a symmetric prediction algorithm, then the jackknife+ interval satisfies

$$\mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{jack}+}(\mathbf{X}_{n+1}) \right) \geq 1 - \alpha - 4\sqrt{\nu_n} - 4\varepsilon_n c.$$

Theorem (Asymptotic training conditional coverage for jackknife - Steinberger & Leeb (2023), Amann et al. (2023))

If $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$ are i.i.d., under Assumptions $(\mathcal{AC}^b)$, $(\mathcal{SA}_2)$ and a boundedness assumption on the prediction error, if $\eta_n \rightarrow 0$ then the jackknife interval satisfies

$$\lim_{n \rightarrow \infty} \mathbb{E} \left\{ \left| \mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}^{\text{jack}}(\mathbf{X}_{n+1}) | \mathcal{D}_n \right) - (1 - \alpha) \right| \right\} \rightarrow 0.$$

Resampling and training cond. cov.: research questions

- ▶ What if the samples come from a design of experiments? (standard practice in engineering)
- ▶ Computational bottleneck: can we leverage e.g. the infinitesimal jackknife to approximate leave-one-out predictions, achieving jackknife+ guarantees without the prohibitive retraining cost?

Outline

Context, introduction to conformal prediction

Resampling and training conditional coverage

Distribution shift

Adaptivity and test conditional coverage

Distribution shift

Exchangeability fails if training and test data follow different distributions. Addressing this problem in all generality is a hard task, this is why we focus here on one specific instance that is of particular interest in practice: the *covariate shift* setting.

In this case we assume that the marginal distribution of the features (or covariates) $\mathbf{X}$ differs in training and test data, but the conditional distribution of $Y|\mathbf{X}$ is the same. In other words, we have $P_{\mathbf{X}Y}^{\text{train}} = P_{\mathbf{X}}^{\text{train}} \times P_{Y|\mathbf{X}}$ and $P_{\mathbf{X}Y}^{\text{test}} = P_{\mathbf{X}}^{\text{test}} \times P_{Y|\mathbf{X}}$.

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Interestingly, in this case, it is possible to perform **weighted conformal prediction** by using quantiles on a weighted empirical distribution of the data and still guarantee marginal coverage. We compute quantiles on a *weighted* empirical distribution of the residuals:

$$\hat{q}_{n,\alpha}^w = \text{Quantile}_{1-\alpha} \left(\frac{1}{\sum_j w(\mathbf{x}_j)} \sum_{i=1}^n w(\mathbf{x}_i) \delta_{R_i} + \frac{w(\mathbf{x}_{n+1})}{\sum_j w(\mathbf{x}_j)} \delta_{+\infty} \right)$$

where the weight function $w = dP_{\mathbf{X}}^{\text{test}}/dP_{\mathbf{X}}^{\text{train}}$ (the likelihood ratio) adjusts the interval width dynamically.

Distribution shift in action

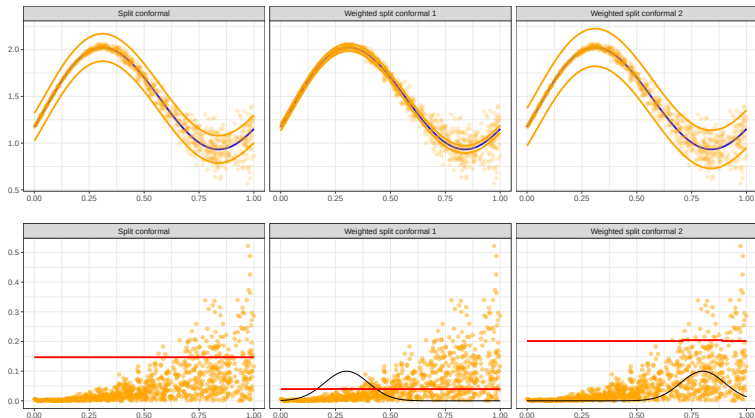


Figure: Bottom row: calibration data residuals (orange dots), quantile value (red line) and distribution of the shifted covariate (black line).

Distribution shift: active learning

For weighted conformal prediction, we emphasized that the shifted distribution of the covariates $P_{\mathbf{X}}^{\text{test}}$ must be known in advance.

Although this is the case in some examples, a more representative setting of practical applications would be that, if we consider $P_{\mathbf{X}}^{\text{test}}$ as a region of interest where we want to guarantee coverage, such a region will most often come from insights gained from the trained predictor itself.

This could be feature regions where the target is predicted to be close to a threshold (for reliability studies) or where it is small (for minimization problems), among others.

Distribution shift: active learning

In this setting, this means that $P_{\mathbf{X}}^{\text{test}}$ is not given beforehand, but is deduced from the training data (through the predictor), and we can formally write

$$P_{\mathbf{X}Y}^{\text{train}} = P_{\mathbf{X}}^{\text{train}} \times P_{Y|\mathbf{X}}, \quad P_{\mathbf{X}Y}^{\text{test}} = P_{\mathbf{X}; \mathcal{D}_n}^{\text{test}} \times P_{Y|\mathbf{X}}$$

where we made this dependence explicit.

Theoretically such assumptions no longer fall in the weighted conformal prediction framework, but recently it has been shown that actually we can still have coverage in this generalized setting.

Theorem (Training conditional coverage for weighted split conformal intervals based on training data - Fannjiang et al. (2022))

Assume we are in the covariate shift setting above with $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)$ i.i.d. from $P_{\mathbf{X}Y}^{\text{train}}$ which is split in $\mathcal{D}_n^{\text{ptrain}} \cup \mathcal{D}_n^{\text{cal}}$ and denote $P_{\mathbf{X}; \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}} = P_{\mathbf{X}; \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}}$ a shifted covariate distribution which depends on $\hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}$, a predictor trained on $\mathcal{D}_n^{\text{ptrain}}$. For the residuals on the calibration data $R_i = |Y_i - \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}(\mathbf{X}_i)|$, $i \in \mathcal{D}_n^{\text{cal}}$ we consider the weighted split prediction interval

$$\hat{C}_{\mathcal{D}_n}^{\text{split}}(\mathbf{X}_{n+1}) = \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}} \pm \hat{q}_{n,\alpha}^w \{(R_i)_{i \in \mathcal{D}_n^{\text{cal}}} \cup \{+\infty\}\}$$

with the weight function given by $w = dP_{\mathbf{X}; \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}} / dP_{\mathbf{X}}^{\text{train}}$. Then, if $dP_{\mathbf{X}; \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}}$ is absolutely continuous with respect to $dP_{\mathbf{X}}^{\text{train}}$, we have

$$\mathbb{P} \left(Y_{n+1} \in \hat{C}_{\mathcal{D}_n}^{\text{split}}(\mathbf{X}_{n+1}) | \mathcal{D}_n^{\text{ptrain}} \right) \geq 1 - \alpha$$

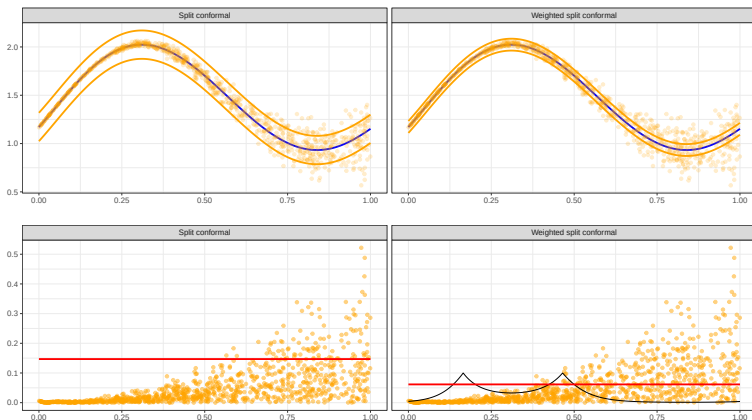
for $(\mathbf{X}_{n+1}, Y_{n+1})$ from $P_{\mathbf{X}Y}^{\text{test}} = P_{\mathbf{X}; \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}} \times P_{Y|\mathbf{X}}$ independently.

Marginal coverage is obtained if $P_{\mathbf{X}; \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}}$ is absolutely continuous with respect to $dP_{\mathbf{X}}^{\text{train}}$ for all pretraining datasets $\mathcal{D}_n^{\text{ptrain}}$.

Distribution shift: active learning

Imagine we want coverage guarantees specifically when the target is close to $t = 1.8$:

$$P_{\mathbf{x}; \hat{\mu}_{\mathcal{D}_n^{\text{pttrain}}}(\mathbf{x})} \propto \exp\left(-\lambda|\hat{\mu}_{\mathcal{D}_n^{\text{pttrain}}}(\mathbf{x}) - t|\right)$$



Notice how the shifted distribution automatically concentrates in the two regions where the predictor is close to the threshold.

Distribution shift: research questions

- ▶ For optimization specifically, recent ideas close to this principle have been proposed for conformalized Bayesian optimization (Stanton et al. 2023, Deshpande et al. 2024)
- ▶ Appart from the specific case of covariate shift, the more general problem of *distribution drift* where the distribution of training data also evolves (with time for example for timeseries) is more tricky, even if recent attempts were made towards general coverage guarantees, see for example Barber et al. (2023), Zaffran et al. (2022)
- ▶ Label shift: weighted conformal prediction elegantly handles covariate shift (P_X), but providing valid intervals when the conditional distribution itself evolves ($P_{Y|X}$) remains a major open challenge

Outline

Context, introduction to conformal prediction

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Adaptivity and test conditional coverage

The quest for adaptivity

Another key aspect of the conformal methods discussed so far is that in general they will tend to produce intervals with constant width (i.e. which does not depend on $\mathbf{x}$):

- ▶ $\hat{C}_{\mathcal{D}_n}^{\text{split}}(\mathbf{x}) = \hat{\mu}_{\mathcal{D}_n^{\text{pttrain}}}(\mathbf{x}) \pm \hat{q}_{n,\alpha}^+ \{R_i, i \in \mathcal{D}_n^{\text{cal}}\}$ from split conformal has constant width by definition
- ▶ The same is true for $\hat{C}_{\mathcal{D}_n}^{\text{jack}}(\mathbf{x}) = \hat{\mu}_{\mathcal{D}_n}(\mathbf{x}) \pm \hat{q}_{n,\alpha}^+ \{R_i, i = 1, \dots, n\}$ from jackknife
- ▶ This is approximately true for

$$\hat{C}_{\mathcal{D}_n}^{\text{jack}+}(\mathbf{x}) = \left[\hat{q}_{n,\alpha}^- \{ \hat{\mu}_{-i}(\mathbf{x}) - R_i^{\text{LOO}} \}, \hat{q}_{n,\alpha}^+ \{ \hat{\mu}_{-i}(\mathbf{x}) + R_i^{\text{LOO}} \} \right],$$

since for predictors with algorithmic stability we know that $\hat{\mu}_{-i}(\mathbf{x})$ will be close to $\hat{\mu}_{\mathcal{D}_n}(\mathbf{x})$, and most of the predictors you use in practice are stable (recall the list in the previous slide)

The quest for adaptivity: the impossibility theorem

Ideally, we want intervals that perfectly adapt to the test point $\mathbf{x}$, satisfying **test conditional coverage**:

$$\mathbb{P} \left(Y_{n+1} \in \hat{\mathcal{C}}_{\mathcal{D}_n}(\mathbf{X}_{n+1}) | \mathbf{X}_{n+1} = \mathbf{x} \right) \geq 1 - \alpha \quad (3)$$

Theorem (Impossibility of test conditional coverage - Vovk (2012), Barber et al. (2021a))

For continuous distributions, any distribution-free procedure satisfying strict test conditional coverage will produce intervals of **infinite expected length**:

$$\mathbb{E} \left[\lambda \left(\hat{\mathcal{C}}_{\mathcal{D}_n}(\mathbf{x}) \right) \right] = \infty$$

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- ▶ Relaxing the conditioning (e.g., binning or local ℓ_2 balls)
- ▶ Relaxing the probability requirement over specific function classes

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To make progress, we must relax our demands:

- ▶ Relaxing the conditioning (e.g., binning or local ℓ_2 balls)
- ▶ Relaxing the probability requirement over specific function classes
- ▶ Let the **score function** used to build the intervals depend on the data distribution (goal: pivotal score function i.e. $\perp \mathbf{X}$)

The quest for adaptivity: score function

Since strict test conditional coverage is impossible, one practical alternative is to design intervals that dynamically adapt their width to local heteroscedastic noise or biased predictive models.

Basic CP methods produce intervals with **constant width** because they rely purely on absolute residuals.

The quest for adaptivity: score function

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A **score function** $S((\mathbf{X}, Y), \mathcal{D}_n)$ maps a data point to a real number indicating nonconformity. By replacing absolute residuals with a generalized score function, split CP retains strict theoretical validity:

$$\hat{C}_{\mathcal{D}_n}^{\text{split}}(\mathbf{x}) = \left\{ y \in \mathbb{R} : S\left((\mathbf{x}, y), \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}\right) \leq \hat{q}_{n,\alpha}^+ \left\{ S_i, i \in \mathcal{D}_n^{\text{cal}} \right\} \right\}$$

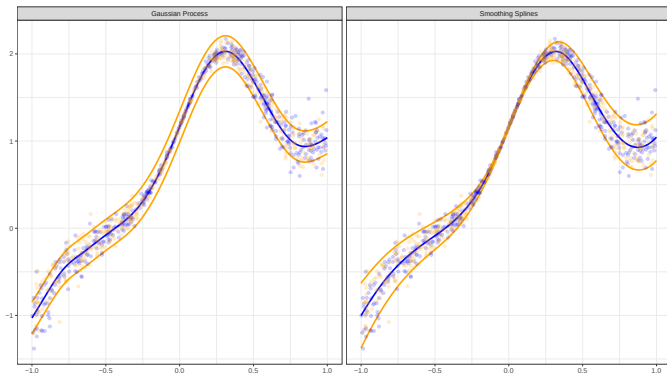
The evolution of CP adaptivity is purely the evolution of choosing smarter score functions.

The quest for adaptivity: rescaled residuals

The simplest path to adaptivity is scaling residuals by an estimator of the local standard deviation $\hat{\sigma}_{\mathcal{D}_n}(\mathbf{X})$:

$$S((\mathbf{X}, Y), \mathcal{D}_n) = \frac{|Y - \hat{\mu}_{\mathcal{D}_n}(\mathbf{X})|}{\hat{\sigma}_{\mathcal{D}_n}(\mathbf{X})}$$

$\hat{\sigma}_{\mathcal{D}_n}$ can be computed explicitly by training a secondary model on the absolute residuals of $\hat{\mu}_{\mathcal{D}_n}$, or inherited from the predictive model itself (e.g. GP).

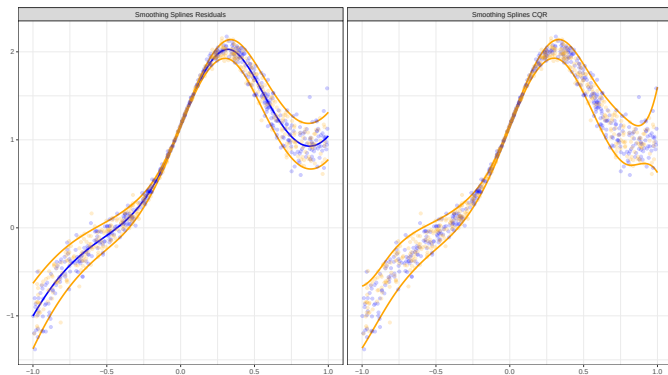


The quest for adaptivity: Conformalized Quantile Regression (CQR)

Instead of estimating the mean and scaling by variance, CQR (Romano et al. 2019) directly estimates conditional quantiles:

$$S((\mathbf{X}, Y), \mathcal{D}_n^{\text{ptrain}}) = \max\left(\hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}^{\alpha/2}(\mathbf{X}) - Y, Y - \hat{\mu}_{\mathcal{D}_n^{\text{ptrain}}}^{1-\alpha/2}(\mathbf{X})\right)$$

This naturally captures asymmetric noise and complex conditional distributions without requiring a two-step variance estimation.



The quest for adaptivity: learning the score function

We just saw that local adaptivity can be greatly improved if we change the score function, and make it depend on estimates of features of the conditional distribution (standard deviation or quantiles).

But intrinsically, such features are learnt in some kind of black-box and unsupervised way, in the sense that they are functions trained on the data without accounting for the main target, that is coverage.

Can we bypass heuristic estimators and directly learn a parameterized score function optimized for **coverage** and **tight average length**?

The quest for adaptivity: learning the score function

As an illustration, for split CP, Xie et al. (2024) consider the score function

$$S(X, Y) = \max(\hat{\mu}_1(X) - Y, Y - \hat{\mu}_2(X)) / \hat{\sigma}(X)$$

inspired by CQR.

The score function is parameterized by three **unknown functions** $(\hat{\mu}_1, \hat{\mu}_2, \hat{\sigma})$ such that $\hat{\mu}_1(\cdot) \leq \hat{\mu}_2(\cdot)$ and $\hat{\sigma}(\cdot) \geq 0$, which are iteratively optimized during boosting rounds with a **task-specific loss function** ℓ .

For example, they propose to minimize the average length

$$\ell(\hat{\mathcal{C}}_{\mathcal{D}_n}) = \frac{1}{n} \sum_{i=1}^n |\hat{\mathcal{C}}_{\mathcal{D}_n}(\mathbf{x}_i)|.$$

The quest for adaptivity: learning the score function

A different point of view was introduced by Liang (2022), Fan et al. (2023), and we recently extended in Allain et al. (2026b).

The starting point is to strongly target adaptivity, by learning a score function which achieves 100% coverage on the pretraining set.

For a score function centered on the mean predictor

$S(\mathbf{X}, Y) = (Y - m(\mathbf{X}))^2 / f(\mathbf{X})$, this translates into the constraint

$$f(\mathbf{X}_i) \geq (Y_i - m(\mathbf{X}_i))^2, \quad i \in [n]$$

Similarly to Xie et al. (2024), we can then write a learning problem to target prediction intervals with small average length, given by $\sum_{i=1}^n f(\mathbf{X}_i)$.

The quest for adaptivity: learning the score function

We formulate a task-specific optimization problem to minimize the average interval length $\sum f(\mathbf{X}_i)$:

$$\begin{aligned} \inf_{m \in \mathcal{F}^m, f \in \mathcal{F}^f} \quad & \frac{a}{n} \sum_{i=1}^n (Y_i - m(\mathbf{X}_i))^2 + \frac{b}{n} \sum_{i=1}^n f(\mathbf{X}_i) + \lambda \|f\| \\ \text{s.t.} \quad & f(\mathbf{X}_i) \geq (Y_i - m(\mathbf{X}_i))^2, \quad i \in [n], \\ & \|m\|^2 \leq s \end{aligned}$$

Learning the score function with kernel sum-of-squares (kSoS)

$f(\mathbf{X})$ must be strictly non-negative. Standard non-negative transformations destroy the convexity of the optimization problem.

Solution: We utilize the **kernel sum-of-squares (kSoS)** framework (Marteau-Ferey et al. 2020). Let $\mathcal{A}$ be a positive semi-definite (PSD) operator in an RKHS $\mathcal{H}$. We define:

$$f_{\mathcal{A}}(\mathbf{X}) = \phi(\mathbf{X})^{\top} \mathcal{A} \phi(\mathbf{X})$$

By construction, $f_{\mathcal{A}}(\mathbf{X})$ is rigorously non-negative.

Theorem (Representer Theorem - Marteau-Ferey et al. (2020))

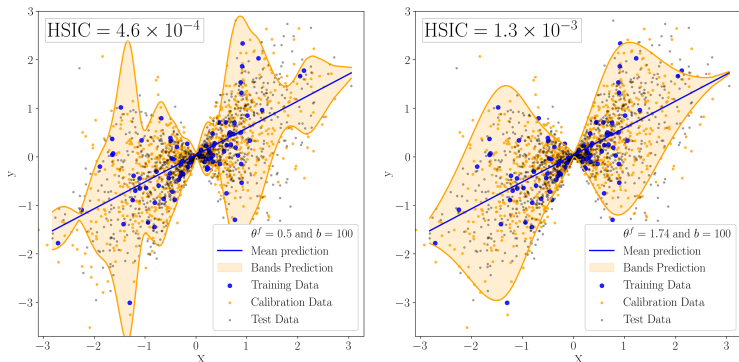
The kSoS optimization admits a finite-dimensional matrix representation:

$$\mathcal{A}^* = \sum_{i,j=1}^n B_{ij}^* \phi(\mathbf{X}_i) \phi(\mathbf{X}_j)^{\top} \Rightarrow f_{\mathcal{A}^*}(\mathbf{X}) = \mathbf{k}_{\mathbf{X}}^{\top} \mathbf{B}^* \mathbf{k}_{\mathbf{X}}$$

This reduces our problem to a convex semi-definite program (SDP) that can be optimized using off-the-shelf solvers.

Learning the score function with kernel sum-of-squares (kSoS)

The lengthscales θ^f inside the kernel dictate the wigglyness of f , which provides direct geometric control over the adaptivity of the prediction bands.



The hyperparameters are optimized to get a score function as independent from $\mathbf{X}$ as possible (pivotal score), with the Hilbert-Schmidt Independence Criterion (HSIC).

The quest for adaptivity: research questions

- ▶ Handling asymmetric noise with automatic adaptation to the data (Allain et al. 2026a)
- ▶ Building approximately pivotal scores (Plassier et al. 2025, Laplante 2026)
- ▶ Decision-theoretic adaptivity: moving beyond simply minimizing average interval length by optimizing score functions for specific downstream asymmetric costs (e.g., the cost of under-predicting is much higher than over-predicting)
- ▶ Benchmarking: criteria to measure adaptivity in real-world settings (Braun et al. 2025)
- ▶ Seeking adaptivity with deterministic outputs? (Ding et al. 2023, Kaur et al. 2025)

Conclusion

- ▶ Guarantees: conformal prediction provides mathematically rigorous uncertainty quantification without heavy distributional assumptions
- ▶ Limits: true test conditional coverage is mathematically impossible for continuous distributions, necessitating a shift in perspective
- ▶ Exciting: many research questions for an operational use in the future!

Thank You!

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